

BGIN Layer 2 Sydney Workshop — PQC Migration & Common Lexicon

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Session summary

APAC satellite workshop extending the Block 14 PQC migration and common lexicon discussions to an Australian audience. The session reprised the core PQC transition landscape presentation and competition announcement from the Tokyo Block 14, then opened to a room that included quantum researchers, IP lawyers, and blockchain practitioners — producing some new angles not surfaced in the Block 14 sessions.

What landed

PQC migration landscape (reprise)

The same foundational presentation from the Block 14 Crypto Agility session was delivered: ECDSA vulnerability, the Falcon key/signature size problem (897 byte key vs. 33 byte ECDSA), the throughput-decentralization tradeoff, migration strategies (ZKP of mnemonic, hybrid signatures, pure PQC, centralized solutions), the governance trilemma (decentralization × security × overhead), stakeholder incentive analysis, and the gap analysis (no definition of PQC-secure blockchain, no migration scope, no incentive-aligned strategy).

Competition structure clarified

The Japanese government competition timeline was presented with more specificity than in Tokyo:

- **2026:** Define specification and evaluation criteria, academic workshop in late October, call for papers
- **2027 (Year 1):** Applicants submit technology documents, software implementations, and self-evaluation. BGIN creates the global testbed (10-30 nodes: North America, Asia, Australia, Europe). Evaluation begins.
- **2028 (Year 2):** Winner selected. Prize pool: ~\$1M USD total. Winner's solution must be publicly available, free of charge — open source requirement for the winning migration method.

Key difference from NIST competitions: NIST provides no prize money; this competition does. The IP policy requires winners to make their solution public, though the competition IP framework is still being finalized.

Bitcoin vs. Ethereum migration divergence

A participant posed the sharpest new question of the session: will Bitcoin and Ethereum take fundamentally different migration paths? The framing: Ethereum's embedded logic and account abstraction flexibility may allow retrofitting PQC into the existing ecosystem (migration of code), while Bitcoin's simpler value-based architecture may push toward bridging value to an already quantum-ready chain (migration of value).

The response acknowledged this divergence: some new chains are positioning as "quantum havens" with EVM compatibility from genesis block, offering a migration destination if Ethereum's in-place migration stalls. However, abandoning a legacy chain means leaving behind all accumulated knowledge, community, and network effects — a significant cost. The Ethereum Foundation's active PQC initiatives and account abstraction flexibility give it more migration options than Bitcoin's more constrained architecture.

The discussion landed on: there are pros and cons between Bitcoin's simple-value approach and Ethereum's flexible-embedded-code approach, and the competition should accommodate both paths.

IP and open source requirements

An IP lawyer in the room flagged the tension between incentivizing innovation (which typically involves proprietary advantage) and the requirement that solutions be deployable on public permissionless networks (which requires open access). The resolution: the winning solution must be open source and free to use, similar to how NIST competitions require applicants to surrender IP claims. The intermediate period during competition evaluation may allow limited disclosure within the working group before full public release.

Testing environment and Australian node participation

A participant with quantum research connections at UNSW asked about the testing methodology. The testbed will be a live distributed network with global node presence. An offer was made to host Australian nodes through local quantum research networks — a concrete step toward the APAC presence the competition needs.

Abandoning legacy vs. preserving value

A significant sidebar emerged: the idea of simply abandoning legacy chains in favor of quantum-ready alternatives. While technically possible, participants noted the difficulty of preserving asset value during such a transition. The community and network effects accumulated on Bitcoin and Ethereum represent enormous intangible value that doesn't transfer automatically to a new chain.

Overlap with active IKP work

1. PQC / Crypto Agility — competition refinement

This session adds the Australian/APAC perspective and quantum research community to the competition stakeholder map. The Bitcoin vs. Ethereum divergence question should inform how the competition structures its evaluation tracks — potentially separate tracks for

UTXO-based and account-based migration approaches. The testbed node offer from UNSW-connected researchers is a concrete action item.

2. Taxonomy of Harms — the “quantum haven” risk

The discussion of new chains positioning as quantum-safe EVM alternatives raises a governance risk: if value migrates to unproven chains during a quantum panic, the ecosystem may trade one set of vulnerabilities for another (untested consensus, immature security, centralization risks). This is a harm category our taxonomy should address — *migration-induced vulnerability transfer*.

3. Proof of Personhood & Agent Duality

The legacy chain abandonment discussion connects to PoP: if identity credentials, reputation, and governance history are tied to a specific chain, migrating to a quantum-safe chain means losing that identity infrastructure unless it's portable. Our PoP work on chain-agnostic personhood credentials becomes a migration enabler.

4. ICBC Brisbane side event alignment

This Sydney workshop validates the APAC appetite for PQC blockchain discussion. The proposed ICBC Brisbane side event (June 2026) should build on the connections made here, particularly the UNSW quantum research network and the Australian blockchain community's engagement with PQC migration.

Pending actions triggered

- **APAC testbed nodes:** Follow up on the UNSW quantum research network offer for Australian node hosting
- **Competition track structure:** Propose separate evaluation tracks for UTXO-based (Bitcoin-family) and account-based (Ethereum-family) migration approaches
- **IP framework:** The competition needs a clear IP policy document before the call for papers — model on NIST's approach but adapt for the open-source requirement
- **ICBC Brisbane connection:** Use contacts from this Sydney workshop to build the invitation list for the June ICBC side event
- **Migration-induced vulnerability:** Add to Taxonomy of Harms as a risk category — value migration to unproven “quantum haven” chains during panic

Compression seal: 🌐🔗🔒 — *The migration path forks at the architecture: value moves one way, code moves another, and the community decides which chain carries the future.*